Frequency-Oriented Efficient Transformer for All-in-One Weather-Degraded Image Restoration

Tao Gao, Yuanbo Wen, Kaihao Zhang, Jing Zhang, Ting Chen, Lidong Liu, and Wenhan Luo, *Senior Member, IEEE*

***Abstract*— Adverse weather conditions, such as rain, raindrop, snow and haze, consistently degrade images in an unpredictable manner, thereby rendering existing task-specific and task-aligned methods inadequate in addressing this formidable problem. To this end, we investigate the application of Transformer in image restoration and introduce an efficient frequency-oriented method called AIRFormer, which is designed to restore weather-degraded images comprehensively and holistically. Specifically, we identify that the initial self-attention mechanism exhibits distinctive properties akin to a low-pass filter. Therefore, we construct a frequency-guided Transformer encoder by incorporating wavelet-based prior information to guide the extraction of image features. Additionally, considering the non-specific frequency characteristics of self-attention in the later stages, we develop a frequency-refined Transformer decoder that incorporates learnable task-specific queries across spatial dimensions, channel dimensions, and wavelet domains. To facilitate the training of our proposed method, we curate a comprehensive benchmark dataset named AIR40K that, encompasses a wide range of challenging scenarios. Extensive experimental evaluations demonstrate the superiority of our AIRFormer over both task-aligned and all-in-one methods across 15 publicly available datasets. Notably, AIRFormer achieves the best trade-off between the inference time and quality of reconstructed image, comparing with existing methods such as TransWeather and Restormer. The source code, dataset and pre-trained models will be available at https://github.com/chdwyb/AIRFormer.**

***Index Terms*— Computer vision, image restoration, adverse weather removal, frequency-oriented transformer.**

I. Introduction

ALL-IN-ONE image restoration (AIR) techniques have emerged as a significant research focus, aiming to reconstruct high-quality images that have been degraded by various weather conditions, including rain, raindrop, snow, and haze. These weather degradations reduce image visibility and drastically affect the performance of certain computer vision algorithms, such as recognition [1], [2], [3], segmentation [4] and tracking [5], [6]. However, single-degraded image restoration plays the leading role currently, and it can be divided into two categories, 1) task-specific ones [7], [8], [9], [10], [11], [12], [13] and 2) task-aligned ones [14], [15], [16], [17], [18]. The former involves designing specialized models for distinct restoration objectives, while the latter constructs a single network that is trained separately to address multiple degradations. In contrast, all-in-one methods employ a unified network architecture to restore degraded images, offering a promising and practical approach to address the diverse challenges associated with weather-degraded image restoration. However, due to the diversity of image with weather degradations, finding a method that can identify commonalities across different weather conditions is a focal point of research in this field.

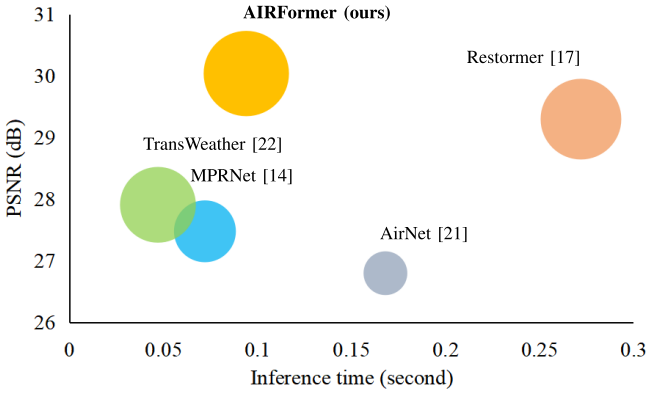


Fig. 1. Efficiency comparison of the evaluated methods for all-in-one weather-degraded image restoration between quantitative metrics and inference time. Our proposed AIRFormer outperforms all other methods [17], [19], [21], [22] and achieves the best trade-off between inference time and image quality.

Recently, there has been a growing interest in the development of all-in-one image restoration methods. One such approach, called All-in-One [19], exploits multiple weather-encoders to accept different degraded images and a sharing decoder to generate restored images, while adopting a discriminator to classify degradations. IPT [20] handles multiple degradations by adopting multiple heads and tails with a pre-trained Transformer. However, the use of separate

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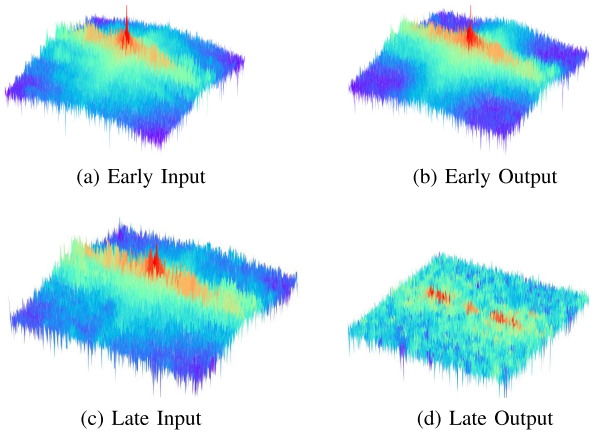
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(a) Early Input (b) Early Output

(c) Late Input (d) Late Output

Fig. 2. **Spectrum variation of the in-/out-put features in self-attention**. The first row demonstrates that early self-attention (SA) exhibits a preference for low-frequency information, while the second row illustrates that late SA operates as a non-specific filtering function. We posit that this behavior can be attributed to the late stages, which not only remove degradations but also reconstruct fine details.

heads and tails in IPT necessitates training them separately from the main body network. In contrast, AirNet [21] and TransWeather [22] adopt an end-to-end scheme to remove unwanted factors during the restoration process. Unfortunately, these methods either suffer from high computational costs or overlook the frequency variation present in the restoration process.

However, most Transformer-based methods [15], [16], [17], [22], [23] primarily focus on improving self-attention (SA) based on spatial features, without adequately considering the knowledge of frequency variation. In this paper, we investigate the general principle of Transformer in terms of image restoration [15], [17] and aim to unify multiple tasks related to adverse weather removal by incorporating frequency changes. We discover that early self-attention exhibits low-pass-like characteristics, while late SA does not function as a prominent filtering mechanism, as depicted in Figure 2. Although different weather degradations may have different frequency band, however their frequency variations of features in the u-shaped architectures are similar, which facilitates our establishment of all-in-one method. Our findings suggest that image restoration networks tend to capture low-frequency information during the early stages and effectively eliminate degradations present in the high-frequency components. Subsequently, the reconstruction network gradually restores high-frequency information to preserve image details while removing degradations. To some extent, this phenomenon explains the use of "pooling" operations and the performance enhancement achieved by SA in the later stages of PoolFormer [24]. With a natural division line between the early and late stages in the U-net architecture [25], we employ a U-shaped PoolFormer as our baseline encoder. Degradations, such as rain streaks [26] and snow particles [27], are often small in scale, leading to difficulties when attending to larger image patches. This limitation hampers the generalization performance without additional supplementary information. To address this issue, we propose a supplementary prior (SP) module that utilizes depth-wise wavelet convolution to fuse features and selectively filter high-frequency components while preserving low-frequency components. The SP module, in conjunction with frequency prior information, facilitates additive guidance for feature learning. Additionally, it enables the Transformer encoder to focus on features within patches after the patch embedding process. Although our method is frequency-oriented, which may not be necessary in the frequency domain for degraded images, it can be equivalently implemented using spatial operations.

Most Transformer-based methods typically define tokens as pixels or patches in the spatial dimension. However, in regard to the unspecific frequency changes in the late stage, it becomes crucial to equally focus on both low and high frequencies to restore images with rich details and accurate regions. By visualizing the feature maps, we find that the spectra of different channels are not exactly the same. Fcanet [28] demonstrates that channel attention primarily enhances low frequencies, indicating that self-attention in the channel dimension emphasizes low-frequency information, while self-attention in the spatial dimension refines frequency at the resolution level. Furthermore, a feature map encompasses both low and high frequencies, implying that more detailed attention is advantageous for image reconstruction and detail preservation. Inspired by [29], we employ a discrete wavelet transformer (DWT) to refine features into frequency components, enhancing feature representation in the frequency domain. In addition, vision Transformer always represents $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ vectors are typically derived from the same input features, resulting in strong dependencies between attention and input features. This characteristic limits the generalization and transferability of the model when dealing with multiple degradations. According to the research in [30] and [22], we introduce special task-biased queries into the frequency refinement decoder, where the $\mathbf{Q}$ performs as prior knowledge related to the consistency of different weather conditions. Meanwhile, most methods [14], [17], [21], [22] primarily focus on the error in the spatial domain while disregarding the error in the frequency domain. This omission hinders the effective enforcement of frequency consistency between the restored and degradation-free images. To address this, we propose a multi-domain consistency loss function to enhance the quality of restored images.

Figure 3 shows our more detailed classification on different methods for weather-degraded image restoration than [21] and [22]. Figure 4 illustrates the overview of our proposed AIRFormer. The main contributions are summarized as follows.

- We initially investigate Transformer from the viewpoint of feature frequencies. Subsequently, we propose AIRFormer as an all-in-one image restoration solution for adverse weather conditions. AIRFormer primarily consists of a frequency-guided Transformer encoder and a frequency-refined Transformer decoder.
- Within our AIRFormer framework, we incorporate a frequency-guided encoder that generates features by leveraging the low-pass-like properties of early self-attention in Transformer. Furthermore, we devise a frequency-refined decoder that refines tokens from the channel dimension, spatial dimension, and wavelet

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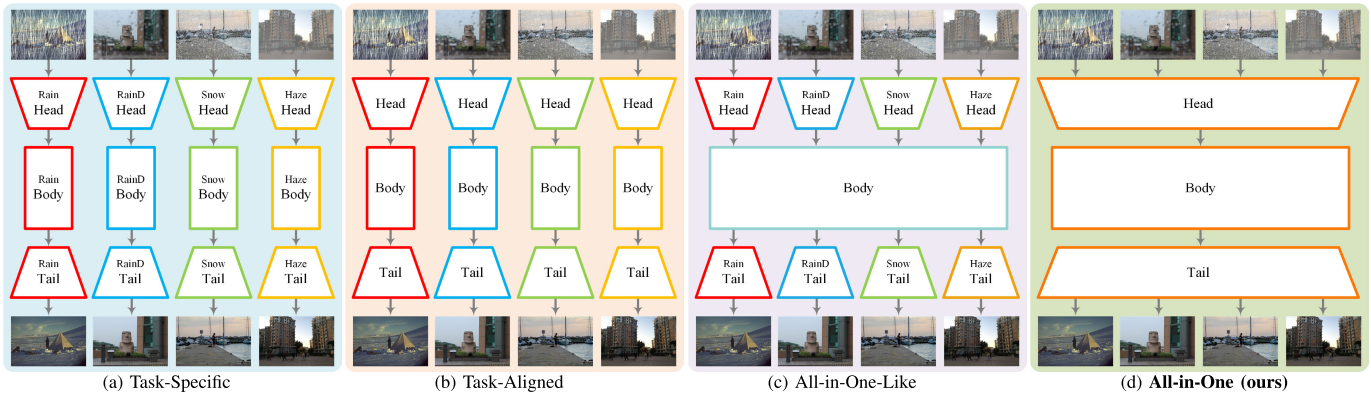


Fig. 3. **Illustrations of different image restoration frameworks**. In early studies, (a) shows task-specific solutions while (b) presents task-aligned networks for restoring images degraded by a single factor. Recently, the all-in-one-like (c) requires different heads and specific tails to achieve multi-degraded image restoration. Our work (d) aims to restore images by traing the exact same network once.

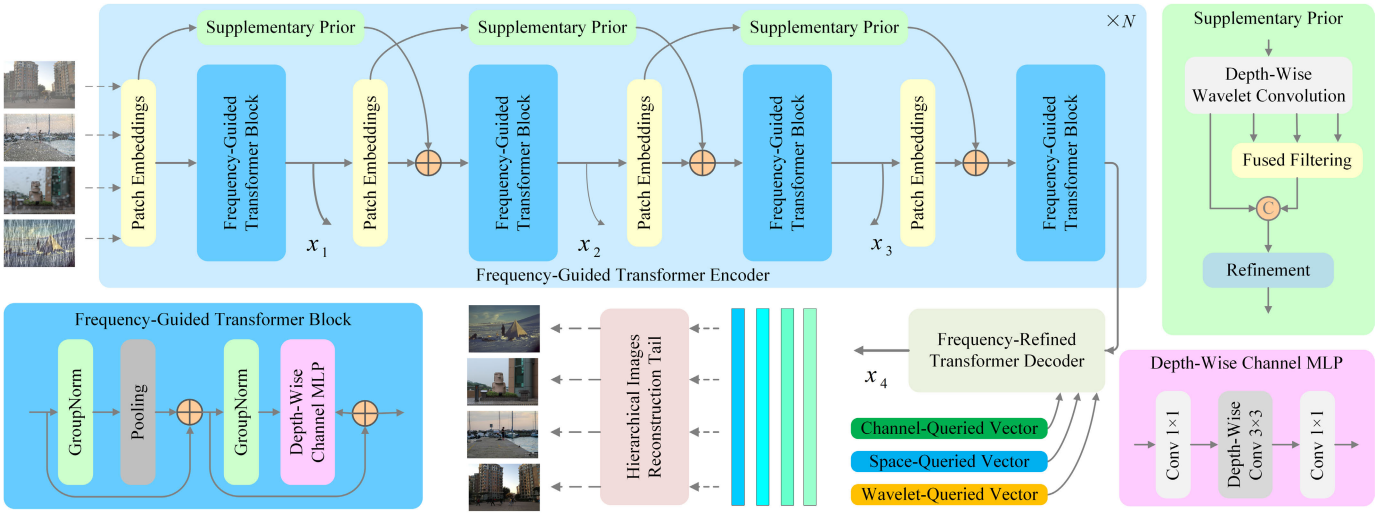


Fig. 4. **Overview of our proposed AIRFormer**. The low-quality images are input into a frequency-guided Transformer encoder through patch embeddings to extract and generate hierarchical features. The learning process is guided by the supplementary prior. Subsequently, a frequency-refined Transformer decoder, which is constrained by spatial, channel, and wavelet queried vectors, generates task-biased features. These encoding and decoding features are then utilized to reconstruct high-quality images.

domain to decode features. These decoding operations are guided by task-biased queries, which aim to enhance the generalization ability of the model.

- Experimental results obtained from the evaluations conducted on 15 public datasets demonstrate that our proposed AIRFormer attains state-of-the-art performance while maintaining a low computational burden.

II. Related Work

A. Task-specific Image Restoration

Early studies establish specialized models for degraded images to characterize the composition of degradation and corresponding ground truth as shown in Figure 3 (a). An image degraded by rain streaks is often modeled as [26], [31], [32]

$$I = B + R_{streak}, \tag{1}$$

where I is the rainy image, B is the rain-free image, and R_{streak} is the rain streaks layer. In the following formulas, we define I as the contaminated image and B as the degradation-free. For instance, DID-MDN [33] estimates rain-density and removes rain streaks using a multi-stream densely connected network. However, it fails to handle the degradation like haze. Liu et al. [27] model the snowy image as

$$I = M \odot S + B \odot (1 - S), \tag{2}$$

where M is a mask indicating snow, $\odot$ denotes element-wise multiplication, S is the snow layer. According to [12], raindrop degenerates image in the way of

$$I = (1 - D) \odot B + R_{drop}, \tag{3}$$

where D is a binary mask and R_{drop} is the raindrop layer. The physical model of an image occluded by fog [34], [35] is

$$I = T \odot B + (1 - T) \odot A, \tag{4}$$

where T is the transmission map produced by scattering effect, A is the scene atmospheric light value. However, the composition and distribution of pollution in reality are different, yet specific modeling limits the solution space and leads to inability to adapt to the diversity [22], [27], [36], [37],

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like rain with various sizes, densities and directions. Lately, CNN and Transformer are introduced to improve the network generalization ability to single-degradation, and achieve great successes over traditional models in single-degradation image restoration tasks, like deraining [26], [33], [37], [38], [39], [40], desnowing [10], [27], [31], [41], [42], raindrop removal [12], [43], [44], [45], [46], [47] and dehazing [48], [49], [50], [51], [52], [53]. A general formula for these task-specific methods is

$$B = f_{\rho,\theta_\rho}(I_\rho)), \tag{5}$$

where f_{ρ,θ_ρ} represents the network model and parameters θ_ρ with respect to degradation type ρ, respectively. As the demand for intelligence is currently increasingly, task-specific methods have been unable to handle a variety of possible degradations. Therefore, seeking a task-aligned solution for different degradations has gradually received attention.

B. Task-Aligned Image Restoration

Although there are inconsistencies between different degradation principles, the powerful representational capability of deep learning leads to task-aligned single-degradation image restoration methods [14], [16], [17], [54], [55], which can be expressed as

$$B = f_{\theta_\rho}(I_\rho), \tag{6}$$

where f_{θ_ρ} denotes the same solution with parameters θ_ρ corresponding to the degraded image I_ρ. Among these methods as shown in Figure 3 (b), [56] presents a multi-level wavelet CNN for enlarging receptive field with better trade-off between efficiency and effectiveness. MIRNet [57] establishes a new feature extraction model, which preserves the original high-resolution features while maintaining accurate spatial details. Later, a style of residual connection is presented, dubbed "dual residual connection" [58], aiming to exploit the potential of paired operations. CLEARER [59] proposes a multi-resolution searching space to find suitable architectures and finally achieve promising performance in denoising and deraining. Recently, MPRNet [14] designs a multi-stage network that produces context-rich and spatially accurate output to achieve progressive restoration, and HINet [60] similarly achieves competitive performance with less computation and time-consumption based on half instance normalization. Along with the development of vision Transformer, SwinIR [15], Restormer [17] and Uformer [16] achieve quite good performance on several image restoration tasks. At the near present, MAXIM [18] conducts a general image processing framework based on multi-axis MLP. Although task-aligned methods present effective generalization and superior performance, the separate training and long-term inference are the stumbling blocks in practical application.

C. All-in-One Image Restoration

Previous methods either establish corresponding models for specific task [12], [26], [27], [48], or learn unified solutions but trained separately [14], [16], [17], [18], without exploring the commonality among different degradations to achieve AIR. There are two formulas for AIR, as shown in Figure 3 (c)

$$B = t_{\rho,\theta_\rho}(f_\theta(g_{\rho,\theta_\rho}(I_\rho))), \tag{7}$$

where t_{ρ,θ_ρ} and g_{ρ,θ_ρ} are the processing tail and head for I_ρ, f_θ are the parameters that remains constant after leaning for all degradations. Please note that t_ρ is not always considered. For example, [19] proposes an AIR method, which contains different acceptance heads for several bad weather conditions. In contrast, IPT [20] develops a pre-trained image processing framework with specific tails for each type of pollution. The other formula is

$$B = f_\theta(I_\rho), \tag{8}$$

where f_θ is self-same for all weather-degraded image restoration tasks as shown in Figure 3 (d). In related research, [61] proposes a decoupled learning framework to adapt several image restoration tasks, which is similar to late [21], [62]. Besides, the method named TransWeather [22] is designed for multi-degraded image restoration with only one single encoder and decoder. Therefore, the all-in-one methods formulated as (8) perform as the promising and effective approaches to eliminate multiple weather degradations unitedly.

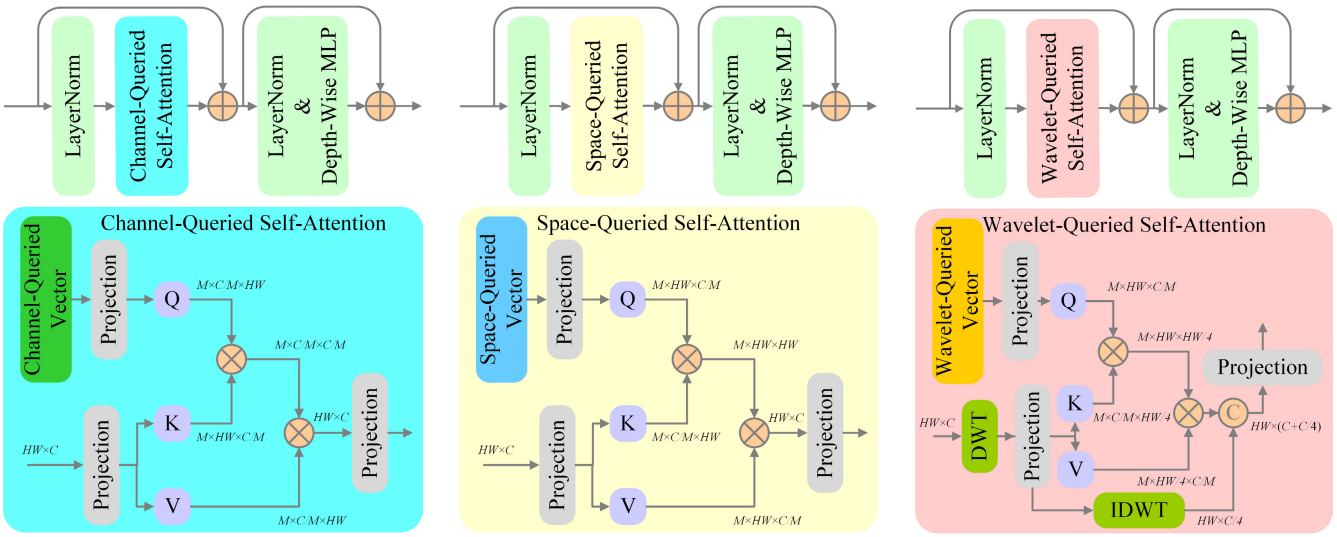
D. Vision Transformer

Transformer has been widely adopted in up-/low-stream computer vision tasks [16], [17], [30], [63], [64], [65], and shown considerable competitiveness. Current works on Transformer believe that spatial SA can effectively learn long-range dependencies and adapt to the input, therefore focus on improving the token mixers, like window partitioning [64], [66], relative position coding [67], convolution combining [68], [69], [70] and formula optimizing [71], [72]. Beyond above, some other works [73], [74], [75] simply replace SA with MLP-like modules and still achieve comparable effects, which questions the dominance of attention-based architectures and inspires the work on what plays the core role in Transformer. For example, PoolFormer [24] applies a simple operator named "*pooling*", which outperforms the mainstream vision Transformer-based methods [64], [74], [76], [77]. However, few works explore and clarify the frequency characteristics of Transformer in image restoration.

III. Proposed AIRFormer

Our main purpose is to explore the feature frequency varying tendency of vision Transformer in image restoration, thus accordingly design a frequency-oriented efficient Transformer to restore the degraded image under adverse weather conditions. The overall pipeline of our proposed AIRFormer is shown in Figure 4. Notably, although our AIRFormer is frequency-oriented, which processes the weather-degraded image in spatial domain equivalently instead of introducing extra frequency transformation into architectural establishment. The main components in our proposed AIRFormer are illustrated specifically in the following sections.

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(a) The channel-queried Transformer block (b) The space-queried Transformer block (c) The wavelet-queried Transformer block

Fig. 5. **Overview of our proposed frequency-refined Transformer decoder**. We calculate self-attention in the spatial and channel dimensions, as well as in the wavelet domain, by utilizing task-specific vectors associated with each dimension. This operation enables the decoder to selectively attend to frequency information at multiple levels.

TABLE I
COMPOSITION OF AIR40K DATASET FOR ALL-IN-ONE IMAGE RESTORATION UNDER ADVERSE WEATHER CONDITIONS

Task	Dataset	Venue	Previous Train Samples	Previous Test Samples	Previous Test Names	AIR40K's Train Samples	AIR40K's Test Samples	AIR40K's Test Names
Deraining	Rain14000 [78]	*CVPR 2017*	12,600	1,400	—	11,200	2,800	Rain2800
	Rain800 [79]	*TCSVT 2019*	700	100	Test100	700	100	Rain100
	Rain100H [80]	*CVPR 2017*	1,800	100	—	1,800	100	Rain100H
	Rain100L [80]	*CVPR 2017*	200	100	—	0	100	Rain100L
	Rain12000 [33]	*CVPR 2018*	12,000	1,200	Test1200	0	1,200	Rain1200
	Rain12 [81]	*CVPR 2016*	12	0	—	0	12	Rain12
Raindrop Removal	Raindrop [12]	*CVPR 2018*	861	58 249	test-a test-b	861	58 249	RainDA RainDB
Desnowing	Snow100K [27]	*TIP 2018*	50,000	16,801 16,588 16611	Snow100K-L Snow100K-M Snow100K-S	9,000	16,801 16,588 16,611	SnowL SnowM SnowS
	CSD [10]	*ICCV 2021*	8,000	2,000	—	8,000	2,000	Snow2000
Dehazing	ITS [34]	*TIP 2018*	13,990	500	RESIDE-Indoor	4,500	500	HazeIn
	OTS [34]		31,3950	500	RESIDE-Outdoor	4,495	500	HazeOut
	HSTS [34]		0	10	HSTS-synthetic	0	10	Haze10
All-in-One	**AIR40K**	**—**	—	—	—	**40,561**	**57,629**	**—**

A. Frequency-Guided Transformer Encoder

Initially, we explore the frequency characteristics of SA and MLP in the pretrained image restoration networks, SwinIR [15] and Restormer [17]. Surprisingly, we find that early SA is low-pass-like while late SA does not work as an obvious filter function. The approximations of early SA and "*pooling*" in frequency properties further support the appropriate adoption of PoolFormer [24] in our baseline of Transformer encoder. However, adverse weather conditions present both approximations and differences, which causes unguided learning to exacerbate the training difficulty. Existing work rarely embeds the frequency information, while a few turn it into a backbone component instead of a bootstrap module. For example, Fnet [82] replaces self-attention with fast Fourier transform. However, our experiments suggest that this kind of approach leads to worse performance than self-attention-based Transformer in image restoration.

The research in [29] reveals that discrete wavelet transform (DWT) [83] improves the representation ability of vision Transformer. We attribute it to the fact that, DWT down-samples the two-dimensional image into wavelet sub-bands and achieves a fine separation of high/low frequencies. Since the frequency changes significantly in image restoration, the separation and enhancement of frequencies are beneficial. The 2-dimensional DWT is defined as

$$\begin{aligned} W_\varphi[j+1,k] &= h_\varphi[-n] * W_\varphi[j,n]\big|_{n=2\ k,k\le 0}, \\ W_\psi[j+1,k] &= h_\psi[-n] * W_\psi[j,n]\big|_{n=2\ k,k\le 0}, \end{aligned} \tag{9}$$

where φ and ψ denote the approximation and detail function respectively, W_φ and W_ψ are the coefficients, $h_\varphi[-n]$ is the time reversed scaling vector while the $h_\psi[-n]$ is the

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wavelet vector, n and j denote the sampling of vector and the resolution level. Due to the low-frequency preservation in early stage, we perform fused filtering on high-frequency components, and then guide the Transformer encoder to extract and generate feature through a refinement module. We name this guiding frequency knowledge as supplementary prior, which can be formulated as

$$\mathbf{X}_{sup}^{l} = SP_l(\mathbf{X}^{l-1}, \mathbf{X}_{sup}^{l-1}), \tag{10}$$

where $\mathbf{X}_{sup}^{l}$ and SP_l denote the mapping approach containing $DWT(\cdot)$, fused filtering and refinement [24] at current stage, $\mathbf{X}^{l-1}$ is the feature enhanced by $\mathbf{X}_{sup}^{l-1}$. When $l = 1$, $\mathbf{X}^{l-1} = 0$ and $\mathbf{X}_{sup}^{l-1} = 0$. On one hand, our frequency-guided Transformer encoder meets the low-pass-like characteristics of backbone network, and on the other hand SP guides the AIRFormer to converge faster and extract more accurate features. In addition, supplementary prior is additively embedded into the Transformer encoder, which provides frequency-oriented information while ensuring that the main features are not drastically transformed.

B. Frequency-Refined Transformer Decoder

The non-specific filtering characteristics and a fact [24] that the classification accuracy is significantly improved by replacing "*pooling*" with SA in later learning period, suggesting it is worse to apply the same token mixer in our Transformer decoder. Therefore, based on the frequency variation in image restoration, we rethink feature decoder architecture in terms of refining frequency.

Most Transformer-based image restoration methods [16], [17] define token as pixel or patch, aggregating context in spatial dimension. However, there are differences in the spectrum regarding the same set of features. In [28], the channel attention is verified to be the lowest-frequency component of discrete cosine transformer (DCT). Obviously, channel-dimension self-attention is a simple proportional transformation in frequency domain from different channels. Based on this, we refine feature frequencies and transfer attention to channel dimension for realizing weighted frequencies transformation. Therefore, channel attention in either CNN or Transformer is a representation in low-frequency level feature. Another fact is that spatial self-attention is more powerful than channel attention in modeling rich frequency dependencies, thereby we first adopt these two-dimension SA to realize frequency-wise refinement. However, Zheng et al. [84] recently demonstrates that self-attention is less powerful than convolution to capture high-frequency dependencies, which inspires us that high/low frequency separation may improve this disadvantage. Furthermore, we consider that simply calculating SA in both space and channel dimension misses the contribution discrepancy at different frequencies levels in single feature plain. Based on this, we adopt DWT to further refine feature frequencies and represent self-attention mapping in wavelet domain. The computation of SA in spatial dimension, channel dimension and wavelet domain focuses on frequency changes from a more holistic and detailed perspective, thereby facilitates the reconstruction of multi-weather degraded image. Although these main architectures have existed, no research ever unified them in terms of frequency refinement to our best knowledge.

In addition, Transformer always represents self-attention from pre-positioned feature maps, and exhibits over-strong dependencies between self-attention and input features, resulting in the limitations of generalization and transferability. According to [30] and [22], we construct task-biased queries separately for frequency-refined decoder. Different with [22], it constrains the attention by both a set of learnable vectors and input features. We illustrate the task-queried attention blocks in Figure 5, where $\mathbf{Q}$ is not all from input features as usual, but vectors partially representing task knowledge based on the particularity of weather-degraded image restoration. In this approach, the features decoded by our decoder not only employ sufficiently frequency information, but also are directed by the all-in-one task-biased queries.

C. Multi-Domain Consistency Loss

The common loss functions calculate loss in space and ignore the multi-domain consistency. However, similarity in space does not necessarily represent the similarity in frequency. We reduce the image to 1-dimensional continuous signal, where a zero signal shows 0 in both time and frequency domains. The MAE between the two signals in Figure 8 and zero signal are equal, but there appears a significant difference in frequency domain. Obviously, this phenomenon also exists in digital images, thereby we propose a multi-domain consistency loss function, which utilizes discrete cosine transform (DCT) to project an image into a set of cosine components for different 2-dimensional frequencies. We define the loss function in frequency domain as

$$\mathcal{L}_{freq} = \sqrt{\|DCT(\mathbf{I}) - DCT(\mathbf{G})\|^2 + \epsilon^2}, \tag{11}$$

where $\mathbf{I}$ and $\mathbf{G}$ denote restored image and ground-truth respectively, and ϵ is a constant empirically set to 0.001. The smooth $\mathcal{L}_1$ loss is also utilized as $\mathcal{L}_{smooth1}$, which is formulated as

$$L_{smooth1} = \begin{cases} 0.5E^2, & \text{if } |E| < 1 \\ |E| - 0.5, & \text{else} \end{cases}, \tag{12}$$

where $E = \frac{1}{HW}\sum_{i=0}^{H-1}\sum_{j=0}^{W-1}[\mathbf{I}(i,j) - \mathbf{G}(i,j)]$. Eventually, our multi-domain consistency loss function is formulated as

$$\mathcal{L} = \mathcal{L}_{smooth1} + \lambda\mathcal{L}_{freq}, \tag{13}$$

where λ is set to 0.04 following [22].

IV. Experiments

We evaluate our proposed AIRFormer for several multi-degraded image restoration tasks, including deraining, raindrop removal, desnowing and dehazing. The more specific details of datasets, training and visualization are attached in the supplementary materials. In the following tables, we **highlight** and underline the best and second-best quantitative scores.

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TABLE II

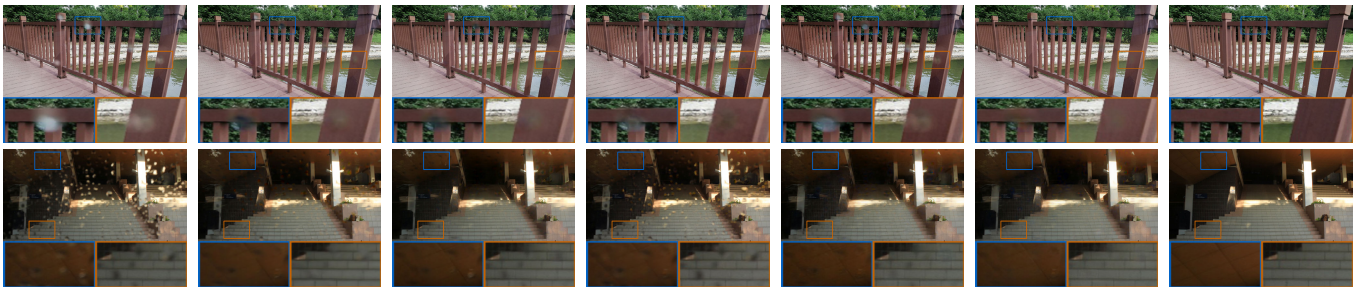
ALL-IN-ONE WEATHER REMOVAL: IMAGE DERAINING. WE TRAIN FOUR STATE-OF-THE-ART METHODS AND OUR AIRFORMER ON AIR40K WHILE TEST ON 6 RAINY DATASETS, THE QUANTITATIVE RESULTS ACHIEVE FURTHER IMPROVEMENTS IN BOTH PSNR AND SSIM

Methods	Rain2800		Rain100		Rain100H		Rain100L		Rain1200		Rain12		Average	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MPRNet [14]	29.89	0.909	24.96	0.872	25.30	0.865	26.99	0.893	29.33	0.882	31.18	0.950	27.94	0.895
Restormer [17]	**30.80**	0.947	26.57	0.899	26.09	0.850	29.12	0.917	30.41	0.900	33.82	0.936	29.47	0.908
AirNet [21]	29.94	0.933	25.74	0.884	24.88	0.869	27.74	0.942	28.62	0.918	30.52	0.964	27.91	0.918
TransWeather [22]	27.88	0.890	24.32	0.903	25.92	0.881	30.79	0.960	30.04	0.935	32.61	0.948	28.59	0.920
AIRFormer	30.62	**0.964**	**26.91**	**0.916**	**26.87**	**0.901**	**32.06**	**0.968**	**30.55**	**0.967**	**34.50**	**0.979**	**30.25**	**0.949**



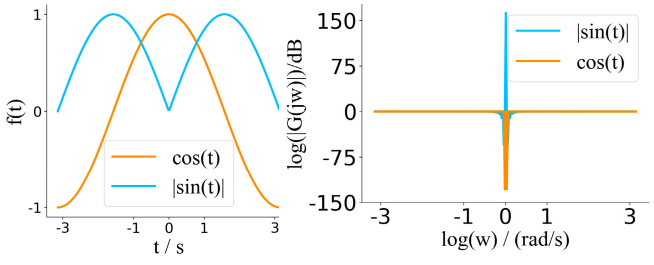
(a) Input (b) MPRNet [14] (c) Restormer [17] (d) AirNet [21] (e) TransWeather [22] (f) **AIRFormer** (g) Ground Truth

Fig. 6. Sampled results of comparative methods and our AIRFormer on all-in-one deraining task, where the color boxes correspond to the zoom-in patches for better visual comparison.



(a) Input (b) MPRNet [14] (c) Restormer [17] (d) AirNet [21] (e) TransWeather [22] (f) **AIRFormer** (g) Ground Truth

Fig. 7. Sampled results of comparative methods and our AIRFormer on all-in-one raindrop removal, where the color boxes correspond to the zoom-in patches for better visual comparison.



(a) The Signals (b) The Spectrums

Fig. 8. Comparison of temporal signals and corresponding spectrums with the same mean square error to zero-like signal in time domain.

TABLE III

ALL-IN-ONE WEATHER REMOVAL: IMAGE DEHAZING. ON BOTH AVERAGE PSNR AND SSIM, OUR AIRFORMER DEMONSTRATES FAVORABLE PERFORMANCE OVER THE OTHER METHODS

Methods	RainDA		RainDB		Average	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MPRNet [14]	27.48	0.903	24.09	0.842	25.79	0.873
Restormer [17]	29.30	0.926	26.10	0.853	27.70	0.890
AirNet [21]	26.80	0.917	23.44	0.825	25.12	0.871
TransWeather [22]	27.91	0.939	23.82	0.849	25.87	0.899
AIRFormer	**30.04**	**0.948**	**26.48**	**0.874**	**28.26**	**0.911**

A. Datasets

There are two all-in-one datasets from All-in-One [19] and AirNet [21]. The former one is not comprehensive and contains few weather datasets, while the latter one is not composed specifically for adverse weather removal. We collect an all-in-one image restoration dataset named AIR40K, including Rain14000 [26], Rain800 [79], Rain100H [80], Rain100L [80], Rain1200 [33], Rain12 [81], Raindrop [43], Snow100K [27], CSD [10], ITS [34], OTS [34], HSTS [34]. We sample these datasets under different adverse weather conditions to compose a comprehensive AIR40K for all-in-one image restoration as shown in Table I. Therefore, our AIR40K dataset is much broader and more comprehensive than currently existing datasets for all-in-one adverse weather

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(a) Input (b) MPRNet [14] (c) Restormer [17] (d) AirNet [21] (e) TransWeather [22] (f) **AIRFormer** (g) Ground Truth

Fig. 9. Sampled results of four methods and our AIRFormer on all-in-one dehazing task, where the color boxes correspond to the zoom-in patches for better visual comparison.

TABLE IV
ALL-IN-ONE WEATHER REMOVAL: IMAGE DEHAZING. ON BOTH AVERAGE PSNR AND SSIM, OUR AIRFORMER DEMONSTRATES FAVORABLE PERFORMANCE OVER THE OTHER METHODS

Methods	HazeIn		HazeOut		Haze10		Average	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MPRNet [14]	29.18	0.971	28.94	0.965	30.90	0.984	29.67	0.955
Restormer [17]	29.40	0.979	29.66	0.980	33.21	0.979	30.76	0.979
AirNet [21]	26.60	0.961	31.03	0.984	31.66	0.988	29.76	0.978
TransWeather [22]	30.62	0.978	31.19	0.972	32.72	0.990	31.51	0.980
AIRFormer	**31.40**	**0.987**	**32.06**	**0.989**	**33.54**	**0.991**	**32.33**	**0.989**

TABLE V
ALL-IN-ONE WEATHER REMOVAL: IMAGE DESNOWING. OUR AIRFORMER ACHIEVES A PERFORMANCE GAIN UP TO ∼1.38 dB IN PSNR

Methods	SnowL		SnowM		SnowS		Snow2000	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MPRNet [14]	27.56	0.913	31.01	0.950	32.54	0.958	28.95	0.963
Restormer [17]	28.15	0.915	31.99	0.957	**33.53**	**0.965**	29.18	0.964
AirNet [21]	28.16	0.922	31.44	0.952	32.79	0.960	29.32	0.956
TransWeather [22]	28.30	0.924	31.74	0.955	32.95	0.962	31.57	0.967
AIRFormer	**29.00**	**0.925**	**32.15**	**0.957**	33.38	0.964	**32.46**	**0.971**

removal. For fair comparison, we have not made any changes to the image number and distribution of all test datasets.

B. Implementation Details

Our proposed AIRFormer is trained on 256×256 random-cropped patches from our collected AIR40K dataset. We exploit the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}) with an initial learning rate of 2×10^{-4}, which are steadily decreased to 1×10^{-7} by the cosine annealing decay [85]. Meanwhile, random horizontal and vertical flips, 90° rotation and MixUp [86] with probability 0.5 perform the data augmentation. When testing, we pad the input images to a multiplier of 128×128 using symmetric padding on both sides. After inference, we crop the padded images back to original size, thus our AIRFormer adapts to all sizes of the degraded images. All the experiments are performed on NVIDIA GeForce RTX 3090.

C. All-in-One Weather Removal

We compare the PSNR/SSIM [87] scores of the proposed AIRFormer with the state-of-the-art task-aligned methods (MPRNet [14], Restormer [17]) and all-in-one methods (AirNet [21], Transweather [22]). All the experimental methods are trained on our composed AIR40K dataset and PSNR/SSIM scores are calculated by using the functions of peak_signal_noise_ratio and structural_similarity from the package of skimage.metrics, which also can be found in our source code.

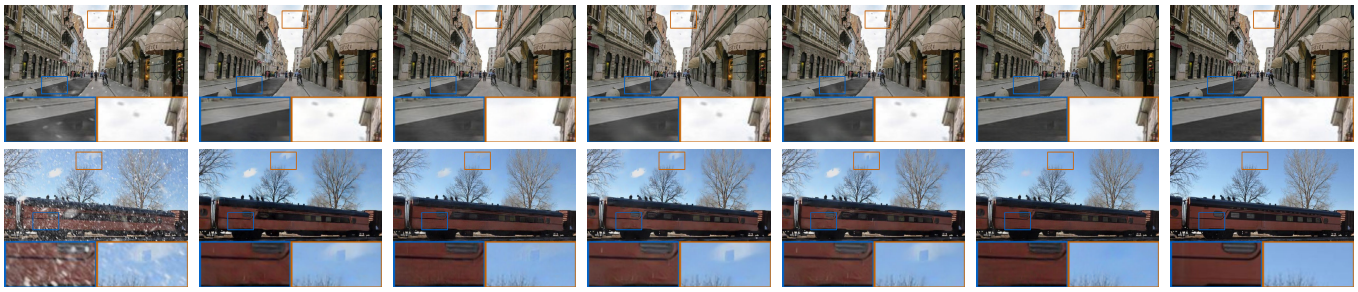
1) Deraining: We report the quantitative performance of evaluated image deraining approaches on 6 rainy test datastes in Table II. Overall, the proposed AIRFormer achieves favorable advantages over the other algorithms. Compared to the recent best task-aligned method Restormer [17], our method obtain average gain of 0.78 dB in PSNR and 4.52% in SSIM. Meanwhile, AIRFormer outperforms the all-in-one TransWeather [22] by 1.66 dB and 3.15% in PSNR and SSIM, respectively. It is notable that AIRFormer performs far ahead in terms of challenging sky restoration as the second row of Figure 6 shown, thereby demonstrating its effectiveness and robustness.

2) Raindrop Removal: For raindrop removal task, we believe that raindrops are similar to snow particles to some extent, thus the training dataset is not extended like [19]. In Table III, we report the PSNR/SSIM scores of several methods on RainDA and RainDB datasets. Our AIRFormer achieves the best performance, for example, it obtains considerable gain of 2.39 dB over TransWeather [22]. Figure 7 shows the visual comparisons on two challenging images. Our AIRformer eliminates raindrops and reconstructs the background finely while other methods tend to result of many black particles.

3) Desnowing: We evaluate our AIRFormer on the test datasets of Snow100K [27] and CSD [10]. Table V shows that our AIRFormer achieves the best performance on SnowL, SnowM and Snow 2000 dataset while the second-best performance on SnowS dataset. Figure 10 illustrates the visual results, where our AIRFormer presents strong generalization capability in removing the snow from sky, which tends to remain after processed by other methods. It can be concluded that AIRFormer is robust to the restoration of low-frequency region and the fidelity of high-frequency part.

4) Dehazing: Table IV shows the quantitative comparisons on image dehazing. Our model surpasses the previous Restormer [17] and TransWeather [22] by 0.59 dB and 1.88 dB

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(a) Input (b) MPRNet [14] (c) Restormer [17] (d) AirNet [21] (e) TransWeather [22] (f) **AIRFormer** (g) Ground Truth

Fig. 10. Sampled results of four methods and our AIRFormer on all-in-one desnowing task, where the color boxes correspond to the zoom-in patches for better visual comparison.

of PSNR, which achieves higher scores on both indoor and outdoor test datasets. We sample and illustrate the results of experimental methods in Figure 9. The dehazed images from our AIRformer attain better global color tone and sharper textures, while maintaining the consistency in flatten region.

D. Single Weather Removal

1) Image Deraining: We choose the datasets of Test1 [88], Rain800 [79] and DID-Data [33] to evaluate our proposed AIRFormer. As Table VI illustrated, AIRFormer obtains the highest SSIM scores on all the datasets and the best performance on the Test1 dataset. On both Rain800 and DID-Data datasets, our proposed AIRFormer achieves the second-best performance in PSNR, which are close to the best Restormer [17].

2) Image Desnowing: The Snow100K dataset is commonly used to evaluate desnowing performance. Table VII shows that our AIRFormer achieves the best performance on Snow100K dataset in both PSNR and SSIM indicators.

3) Raindrop Removal: To evaluate the effectiveness of our proposed AIRFormer on raindrop removal task, we conduct experiments on Test-a and Test-b datasets [12]. Table VIII reports that AIRFormer obtains the best performance in SSIM and the second-best performance in PSNR (where the scores are close to the best MPRNet [14].

4) Image Dehazing: We choose the datasets from NTIRE 2020 [89] and NTIRE 2021 [90] to evaluate the dehazing performance of our proposed AIRFormer. As Table IX shown, our proposed AIRFormer obtains performance gains of 0.86 dB and 0.63 dB over FSDGN [91] in PSNR.

TABLE VI

SINGLE WEATHER REMOVAL: IMAGE DERAINING. OUR AIRFORMER ACHIEVES THE BEST PSNR ON TEST1 [88] DATASET AND BEST SSIM ON ALL THREE DATASETS [33], [79], [88]

Method	Test1 [88]		Rain800 [79]		DID-Data [33]	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
GMM [81]	16.72	0.784	25.71	0.802	25.81	0.834
DDN [78]	22.59	0.861	25.10	0.823	30.97	0.912
RESCAN [92]	24.17	0.889	24.09	0.841	33.38	0.941
PReNet [39]	25.36	0.892	26.61	0.902	33.17	0.948
MSPFN [37]	—	—	27.50	0.876	33.72	0.955
RCDNet [93]	27.14	0.900	28.59	0.914	34.57	0.956
MPRNet [14]	28.03	0.919	28.09	0.891	33.99	0.959
Restormer [17]	28.44	0.926	**29.12**	0.908	**35.29**	0.964
AIRFormer	**29.21**	**0.935**	28.83	**0.925**	34.89	**0.971**

TABLE VII

SINGLE WEATHER REMOVAL: IMAGE DESNOWING. OUR PROPOSED AIRFORMER ACHIEVES THE BEST PERFORMANCE ON SNOW100K [27] DATASET

Method	Snow100K-L		Snow100K-M		Snow100K-S	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DeepLab [94]	21.29	0.775	24.37	0.857	25.95	0.878
RESCAN [92]	26.08	0.811	29.95	0.886	31.51	0.903
DesnowNet [27]	27.17	0.898	30.87	0.949	32.33	0.950
SPANet [95]	23.70	0.793	28.06	0.868	29.92	0.826
JSTASR [41]	25.32	0.808	29.11	0.884	31.40	0.901
DesnowGAN [96]	28.07	0.921	31.88	0.957	33.43	0.964
DDMSNet [42]	28.85	0.877	32.89	0.933	34.34	0.945
AIRFormer	**29.15**	**0.927**	**33.35**	**0.958**	**34.59**	**0.965**

E. Real Application

To evaluate how the AIRFormer works in real-world scenarioa and verify the generalization ability, we conduct additional experiments on the real-world degraded datasets from [27], [34], and [104]. The real-world rainy, hazy and snowy datasets contains 300, 4322 and 1329 images respectively. Due to the raindrop dataset is obtained in real world, we did not conduct real-world experiments for raindrop removal task. We choose two unsupervised metrics, naturalness image quality evaluator (NIQE) [105] and spatial-spectral entropy-based quality (SSEQ) [106], to objectively measure the performance of our AIRFormer and other methods on real-world degraded images. NIQE is based on the construction of a quality-aware collection of statistical features based on natural scene statistic model. These feature s are derived from a corpus of natural undistorted images. SSEQ utilizes local spatial and spectral entropy features on distorted image across multiple distortion categories. The smaller of these two metrics, the higher the quality of the degradation-free images. Table X shows the quantitative results, which illustrates that our AIRFormer achieves either the best or the second-best performance on the real-world degraded images. We also present extensive samples in the web to further verify the better generalization ability of our AIRFormer over those of other comparative methods. Due to space limitation, the visual results will be released in our github page.

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TABLE VIII

SINGLE WEATHER REMOVAL: RAINDROP REMOVAL. OUR AIRFORMER ACHIEVES THE SECOND-BEST PSNR AND THE BEST SSIM ON TEST-A AND TEST-B DATASETS [12]

Method	Test-a		Test-b	
	PSNR	SSIM	PSNR	SSIM
DSC [97]	24.13	0.855	23.13	0.732
pix2pix [98]	26.79	0.864	23.50	0.715
DDN [78]	29.12	0.891	24.52	0.770
attentiveGAN [12]	31.57	0.902	24.92	0.809
DuRN [58]	31.24	0.926	25.32	0.817
MPRNet [14]	**32.15**	0.937	**26.80**	0.851
IDT [99]	31.87	0.931	—	—
AIRFormer	32.09	**0.945**	25.65	**0.868**

TABLE IX

SINGLE WEATHER REMOVAL: IMAGE DEHAZING. OUR PROPOSED AIRFORMER OBTAINS PERFORMANCE GAINS OF 0.86 DB AND 0.28 DB IN PSNR OVER THE EVALUATED METHODS ON THE HAZY DATASETS FROM NTIRE 2020 [89] AND NTIRE 2021 [90], RESPECTIVELY

Method	NTIRE 2020 [89]		NTIRE 2021 [90]	
	PSNR	SSIM	PSNR	SSIM
DCP [100]	12.29	0.411	11.30	0.605
AODNet [52]	13.44	0.413	13.22	0.613
GCANet [101]	17.58	0.594	18.76	0.768
FFANet [102]	18.51	0.637	20.40	0.806
TDN [103]	17.29	0.616	17.73	0.696
FSDGN [91]	19.99	0.731	20.05	0.786
AIRFormer	**20.85**	**0.740**	**20.68**	**0.832**

TABLE X

REAL-WORLD IMAGE RESTORATION RESULTS. OUR AIRFORMER ACHIEVES EITHER BEST OR SECOND-BEST PERFORMANCE ON REAL-WORLD WEATHER REMOVAL OVER THOSE OF OTHER COMPETING METHODS

Method	Rain		Haze		Snow	
	NIQE	SSEQ	NIQE	SSEQ	NIQE	SSEQ
MPRNet [14]	4.51	28.43	6.12	37.16	3.81	27.88
Restormer [17]	4.36	27.97	5.48	36.69	3.69	27.97
AirNet [21]	**4.06**	27.84	5.86	36.75	3.62	27.49
TransWeather [22]	4.23	28.11	5.31	**36.22**	3.68	26.94
AIRFormer	4.17	**27.39**	**5.18**	36.50	**3.59**	**26.44**

V. ABLATION STUDIES AND DISCUSSIONS

For Ablation study, we conduct the experiments of individual components by training on AIR40K and testing on Rain2800 [26] datasets. Meanwhile, we further discuss and analyze the efficiency and the limitations of our AIRFormer.

A. Individual Components

In this part, we start with a U-shaped PoolFormer [24] as a baseline. Then, we convert the "*pooling*" with self-attention and add the supplementary prior in Transformer encoder. Furthermore, the frequency refinement is utilized to replace the common spatial-dimension attention, while the task-queried vectors perform as the replacements of usual **Q** in vision Transformer. Please note that we conduct experiments on the 4 weather-degraded datasets to analyze where the main improvements come from. The metrical scores are the average indicators of the involved datasets of corresponding weather conditions. Table XI reports the results of ablation study in individual components. It demonstrates that each component contributes to the improved performance. Furthermore, we also conduct ablation experiments on the frequency refinement in Transformer decoder. Table XII shows the comparative results obtained by different permutations of SA in spatial-dimension, channel-dimension and wavelet domain. We experimentally find that the order of [Channel, Space, Wavelet] achieves the best scores, which meets the research in [107] and the demand of feature frequency refinement.

TABLE XI

THE ABLATION EXPERIMENTS OF INDIVIDUAL COMPONENTS IN THE PROPOSED AIRFORMER

Component	Rain	Raindrop	Haze	Snow
	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM	PSNR/SSIM
Baseline	25.18/0.860	24.39/0.838	28.99/0.918	27.58/0.879
+ Decoder Self-attention	27.26/0.879	26.41/0.870	29.78/0.929	29.65/0.900
+ Supplementary Prior	29.18/0.902	27.73/0.877	31.02/0.950	30.46/0.917
+ Frequency Refinement	29.63/0.921	28.00/0.897	31.86/0.962	31.24/0.931
+ Task-queried Vectors	30.25/0.949	28.26/0.911	32.33/0.989	31.75/0.954

TABLE XII

ABLATION EXPERIMENTS OF THE FREQUENCY REFINEMENT IN DECODER OF AIRFORMER

Decoder	PSNR	SSIM
Baseline	26.37	0.862
[Channel, Channel, Channel]	24.79	0.831
[Space, Space, Space]	29.33	0.915
[Space, Channel, Wavelet]	29.96	0.928
[Channel, Space, Wavelet]	**30.52**	**0.944**

TABLE XIII

EFFICIENCY ANALYSIS OF THE FOUR STATE-OF-THE-ART METHODS AND OUR AIRFORMER

Methods	Avg. time (s)	Par.$\times 10^6$	FLOPs$\times 10^9$
MPRNet [14]	0.072	**3.637**	148.6
Restormer [17]	0.282	26.13	141.0
AirNet [21]	0.168	8.930	302.1
TransWeather [22]	**0.047**	38.05	6.132
AIRFormer	0.086	58.34	**5.746**

B. Efficiency Analysis

: We compare our AIRFormer with four methods processing an image of 256 $\times$ 256 in terms of average inference time (Avg. time), trainable parameters (Par.) and floating point operations (FLOPs). Table XIII shows that our method is with the minimum calculation complexity. Meanwhile, we illustrate the efficiency comparison in Figure 1, which shows that our proposed AIRFormer achieves the best tradd-off between image quality and processing time. Overall, AIRFormer is a real-time applicable methods for all-in-one weather removal.

C. Loss Functions

We conduct ablation study to determine the effective and competence of frequency-oriented design. We compare

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TABLE XIV
COMPARISON OF DIFFERENT LOSS FUNCTIONS. OUR MULTI-DOMAIN CONSISTENCY LOSS FUNCTION PERFORMS THE BEST OVER OTHER COMMON USED LOSS FUNCTIONS

Loss Function	PSNR	SSIM
$\mathcal{L}_1$	30.15	0.903
$\mathcal{L}_2$	28.21	0.885
$\mathcal{L}_{smooth1}$	29.53	0.912
$\mathcal{L}_{freq}$	28.74	0.890
$\mathcal{L}_{smooth1} + \mathcal{L}_{freq}$	**30.52**	**0.944**

proposed loss function with several common used losses. Table XIV shows that our loss function achieves the best performance on both PSNR and SSIM. Specifically, our consistency loss function increases 0.37 dB at PSNR over L_1. Although $L_{smooth1}$ obtains lower performance than L_1, we experimentally find that our AIRFormer constrained by $L_{smooth1}$ is easier to converge to fine performance than, thus we adopt $L_{smooth1}$ loss function and frequency domain loss L_{freq} to train our AIRFormer.

D. Limitations

We exquisitely design the AIRFormer through exploring the tendency of features frequency changing in the well-tuned SwinIR [15] and Restormer [17], while the specific reason of frequency changing is not discussed in detail. Moreover, our work focuses only on the four common weather conditions without considering more weather type degradations, like dust, dark light, *etc.*

VI. CONCLUSION

In this work, we explore Transformer in feature frequencies and find that the early SA performs low-pass-like while the later is unspecific. Our discoveries verify the improvement of PoolFormer with "*pooling*" and attention. Furthermore, we build a hierarchical frequency-guided encoder to extract and generate features from degraded images, which is supplemented with the frequency of original patches. Meanwhile, we refine the feature frequencies and represent attention queried by the task-biased vectors from channel dimension, spatial dimension and wavelet domain. In addition, we also formulate a multi-domain loss function to keep the consistency of restored image in spatial and frequency domains. Extensive experiments on both synthetic and real-world multi-degraded images shows that our AIRformer achieves real-time performance while outperforms the current state-of-the-art methods.

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